

3 **Dust–Cloud Vertical Configurations Influence the Effective Radius of Low-Level Warm**
4 **Clouds over Marine and Continental Environments**

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Text S1. Identifying dust and cloud layers using satellite observations

To provide an independent comparison with the ground-based ARM observations, we constructed a satellite-based dust–cloud collocation dataset using observations from the Cloud-Aerosol Lidar and Infrared Pathfinder Satellite Observations (CALIPSO), CloudSat, and the Moderate Resolution Imaging Spectroradiometer (MODIS) collected from June 12, 2006 to December 31, 2017. This period corresponds to the longest overlapping record among the three satellite datasets. These satellite products provide complementary information for characterizing dust–cloud relationships. CALIPSO provides vertically resolved aerosol profiles from spaceborne lidar observations. Its profiles are sampled along track at approximately 0.74-s intervals, with a vertical resolution of 60 m. In this study, we used the CALIPSO pure-dust extinction profile dataset developed by Proestakis et al. (2024) to characterize the vertical structure of dust from satellite observations. This dataset separates dust from mixed aerosol types using CALIOP depolarization information, making it suitable for identifying dust-layer boundaries and calculating dust-layer optical depth (DLOD) (Han et al., 2022). It has also been widely used in studies of three-dimensional dust transport (Gandham et al., 2022) and dust radiative effects (T. Wang et al., 2020). CloudSat provides cloud vertical structure from cloud-profiling radar observations. The product identifies up to 10 cloud layers and provides cloud base height (CBH), cloud top height (CTH), and temperature information for each layer. Each atmospheric profile is sampled every 0.16 s, with a vertical resolution of approximately 240 m (Mace & Zhang, 2014). MODIS retrieves cloud microphysical properties, including cloud droplet effective radius (R_e) and liquid water path (LWP), from passive radiance measurements (Platnick et al., 2017). Because MODIS is a cross-track passive imager, its observations are provided as two-dimensional swath pixels.

To construct the satellite-based dust–cloud collocation dataset, we matched CALIPSO, CloudSat, and MODIS observations in both space and time (Stephens et al., 2018; Winker et al., 2010). To facilitate comparison with the ground-based ARM observations, we retained only satellite overpasses within a $5^\circ \times 5^\circ$ domain centered on the ENA and SGP sites (Fig. 1a in the main text and S1a). Although smaller $1^\circ \times 1^\circ$ and $3^\circ \times 3^\circ$ spatial windows may better represent the local environment near each ARM site, they yielded insufficient samples after cloud screening, dust screening, and dust–cloud vertical-configuration classification to support stable grouped statistics and regression analyses. We therefore used the $5^\circ \times 5^\circ$ domain as the satellite sampling region (Fig. S1a), an approach commonly used in satellite–site collocation studies (McErlich et al., 2021). We collocated the different satellite products along the satellite track using a nearest-neighbor approach (Fig. S1b). Specifically, each CALIPSO profile was used as the reference profile to characterize the vertical structure of dust aerosols. For each CALIPSO profile, we selected the five nearest CloudSat profiles based on the shortest great-circle distance to characterize cloud vertical structure and cloud-top thermodynamic conditions, including CBH, CTH, cloud-layer number, and cloud-top temperature. We also selected the three nearest MODIS pixels for each

CALIPSO profile to characterize cloud microphysical properties, including R_e and LWP. Figure S1b shows an example of CALIPSO–CloudSat–MODIS spatial collocation during a representative satellite overpass over the ENA region, in which the CALIPSO profile serves as the reference sample and the neighboring five CloudSat profiles and three MODIS pixels provide matched cloud vertical-structure and microphysical information (see black-edged markers in Fig. S1b). The same spatial filtering and nearest-neighbor collocation procedure was applied to the SGP region. This satellite-to-satellite collocation approach has been widely used in previous studies (Sporre et al., 2016; Zheng et al., 2025).

After CALIPSO–CloudSat–MODIS collocation, we applied cloud-screening criteria consistent with those used for the ground-based observations. To reduce uncertainty in matching dust layers with clouds under multilayer-cloud conditions, we retained a CALIPSO sample only when all five nearest CloudSat profiles were identified as single-layer clouds. We then calculated the mean CTH and mean cloud-top temperature from these five CloudSat profiles and retained only low-level warm-cloud samples with mean CTH below 3 km and mean cloud-top temperature above 273.15 K. These criteria ensured that the satellite and ARM samples were comparable in terms of cloud type. This collocation procedure yielded 720 and 316 satellite-based dust–cloud collocation samples at ENA and SGP, respectively. The final dataset includes dust vertical structure, DLOD, cloud geometric properties, and R_e .

Text S2. Selection of Meteorological Variables

Given that the meteorological modulation of dust–cloud interactions may vary with the relative vertical arrangement of dust and cloud layers, candidate meteorological variables were selected from different altitude levels relevant to the dust–cloud system to characterize the multilayer environmental background affecting cloud microphysical responses as comprehensively as possible. These candidate variables include thermodynamic and dynamic factors from the near-surface layer, the cloud-adjacent layer, and the overlying atmosphere. The initial variable set consisted mainly of relative humidity, temperature, wind speed, lower-tropospheric stability, and vertical velocity, along with their counterparts at several representative vertical levels.

To assess the correlations and potential redundancy among these candidate variables, correlation matrices were calculated separately for the ENA and SGP sites (Fig. S2). The results show that some thermodynamic variables are strongly correlated, particularly temperature, humidity, and stability metrics at different vertical levels, indicating substantial covariance in the vertical thermodynamic structure of the atmosphere. In contrast, several dynamic variables show relatively weak correlations with thermodynamic variables, suggesting that they may provide more independent environmental information. This initial analysis indicates that, although candidate variables from different vertical levels can represent the meteorological background from multiple

perspectives, some degree of information overlap remains, and subsequent variable selection should therefore balance statistical independence with physical relevance.

On this basis, the final selection of meteorological variables was not determined solely by statistical correlations, but was further guided by the vertical dependence of cloud microphysical processes and the physical meaning of each variable. Because the focus is on low-level warm clouds, whose formation and evolution are primarily influenced by the atmospheric environment at and near cloud level, only representative meteorological layers consistent with the height range of low-level warm clouds were retained to avoid unnecessary redundancy and maintain physical consistency in variable definition and interpretation.

Specifically, relative humidity and temperature were averaged over 750–850 hPa; vertical wind shear was defined as the zonal wind speed difference between 725 and 925 hPa; vertical velocity was averaged over 750–850 hPa; and lower-tropospheric stability was defined as the difference between potential temperature at 700 hPa and near the surface. These variables were used to represent the key thermodynamic and dynamic background conditions influencing the microphysical response of low-level warm clouds (Chakraborty et al., 2018; Jeong et al., 2023; Zhang et al., 2009), and to provide physically consistent meteorological constraints for the subsequent analysis of ACI variations under different dust–cloud vertical configurations.

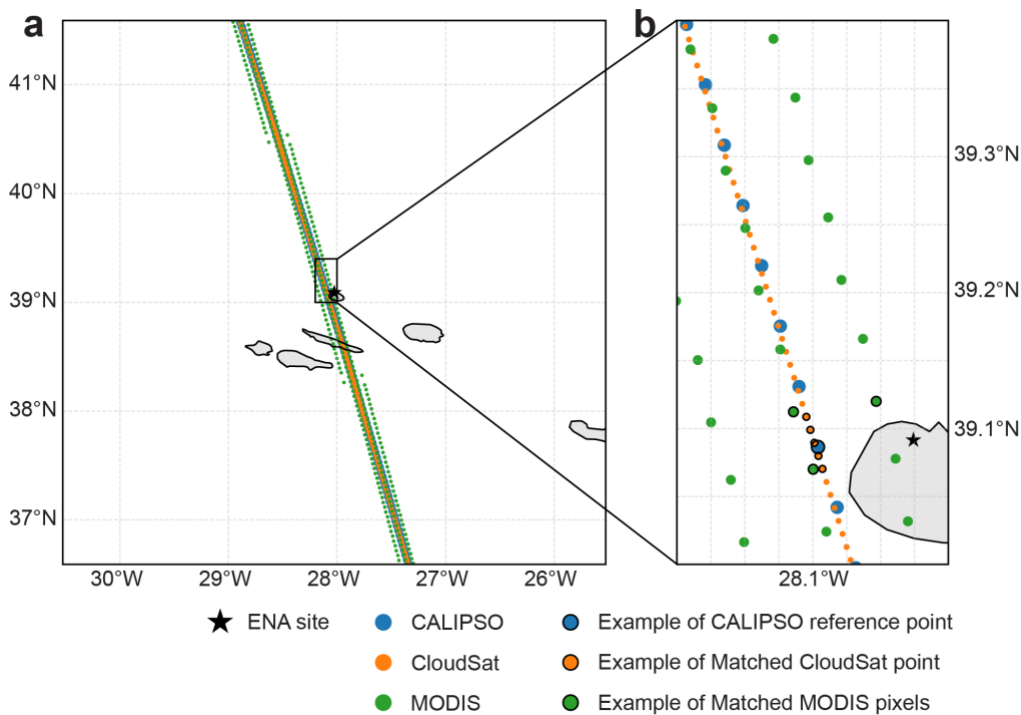


Figure S1. Example of CALIPSO–CloudSat–MODIS spatial collocation over the ENA region. (a) Satellite sampling tracks within the 5° × 5° ENA domain. (b) Enlarged view of one collocation

example, where a CALIPSO reference profile is matched with its five nearest CloudSat profiles and three nearest MODIS pixels.

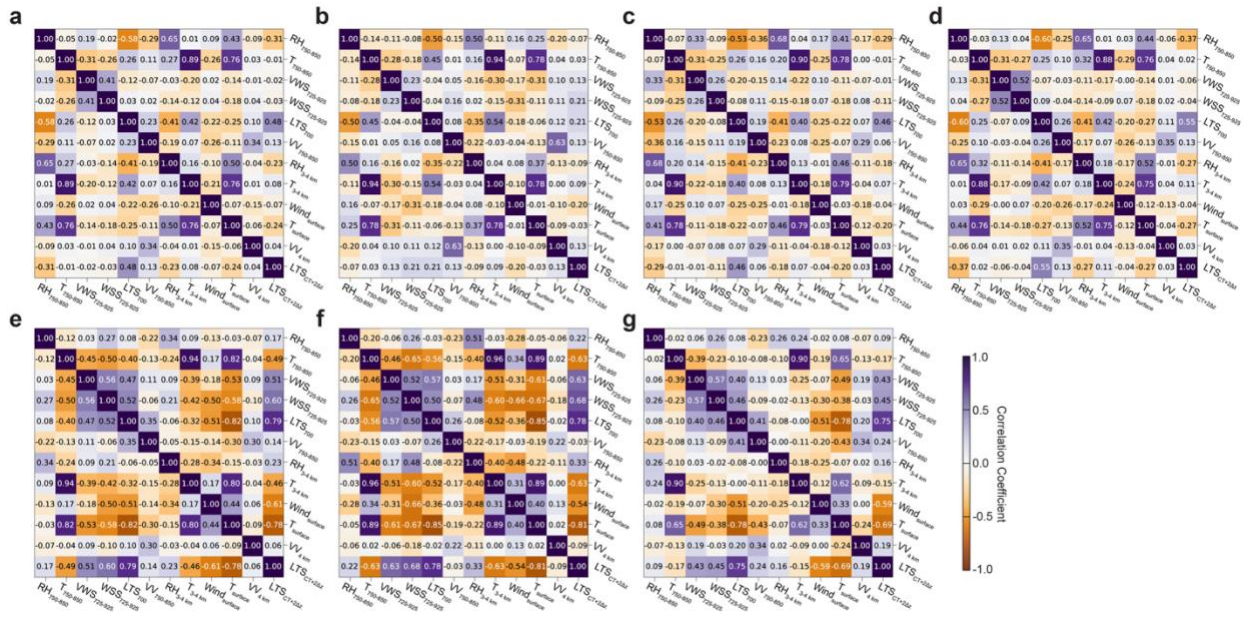


Figure S2. Pearson correlation matrices of candidate meteorological variables. Panels show correlations for ENA total, Top, Middle, and Bottom samples (a–d), and for SGP total, Middle, and Bottom samples (e–g). The candidate variables include relative humidity averaged over 750–850 hPa ($RH_{750-850}$), temperature averaged over 750–850 hPa ($T_{750-850}$), vertical wind shear between 725 and 925 hPa ($VWS_{725-925}$), wind-speed shear between 725 and 925 hPa ($WSS_{725-925}$), lower-tropospheric stability defined using 700 hPa and the near-surface level (LTS_{700}), vertical velocity averaged over 750–850 hPa ($VV_{750-850}$), relative humidity averaged over 3–4 km ($RH_{3-4 \text{ km}}$), temperature averaged over 3–4 km ($T_{3-4 \text{ km}}$), surface wind speed ($Wind_{surface}$), surface temperature ($T_{surface}$), vertical velocity at 4 km ($VV_{4 \text{ km}}$), and lower-tropospheric stability defined using the cloud-top height plus two vertical-resolution levels ($LTS_{ct+2\Delta z}$).

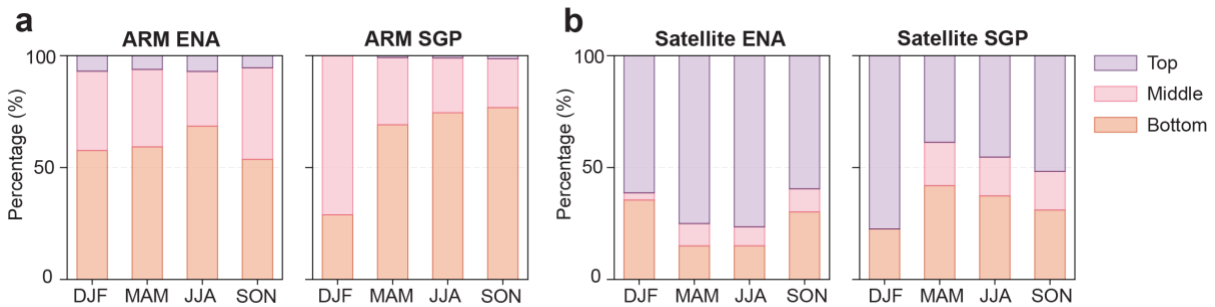


Figure S3. Seasonal occurrence fractions of dust–cloud vertical configurations over ENA and SGP. (a) ARM-derived fractions of Top, Middle, and Bottom configurations at ENA and SGP. (b) Corresponding fractions derived from collocated satellite observations.

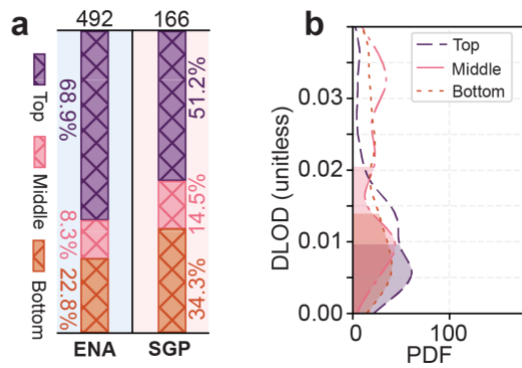


Figure S4. Satellite-derived dust-cloud vertical configurations and dust-layer optical depth (DLOD) distributions. (a) Fractional occurrence of Top, Middle, and Bottom configurations at ENA and SGP, with total sample numbers shown above the bars. (b) DLOD probability density functions for each configuration at ENA. Shading denotes DLOD values below the median.